Market Equilibria under Adverse Selection

Decentralization: the Wilson Doctrine

Can informationally constrained efficient allocations be decentralized through a universal, detail-free system preserving anonymity ?

Wilson, (2020): " It was hard to give up the beautiful welfare theorems implied by "perfect competition" to be able to take account of a welter of detail about agents' incentives in imperfect markets. At this point, we still have to understand if any decentralization is possible at all"

Adverse Selection: a (Neo-)Walrasian Approach?

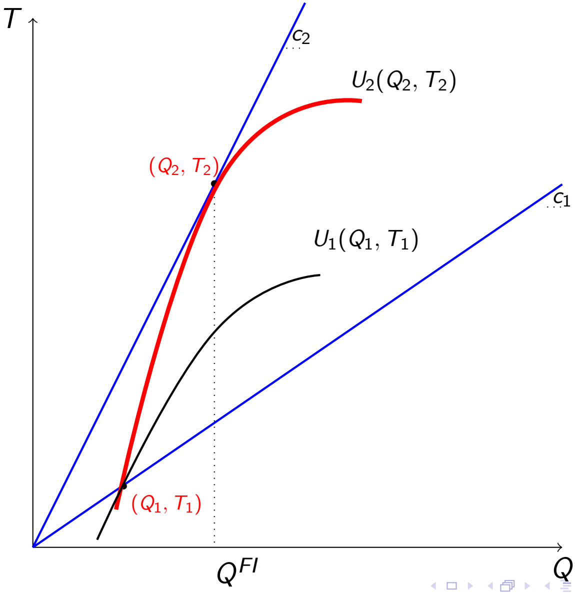
Prescott-Townsend (1984): reformulate the theorems of Welfare Economics by *restricting* to *incentive compatible allocations*

Individuals trade at type-dependent prices their contingent consumption: $c_2 > c_1 \Longrightarrow$ all types want to buy at $c_1 \Longrightarrow (IC)$ may be required for existence

(IC) imposes a **quantity constraint**. Relating the trades made by the different types of an individual, it also generates an **externality**

Under common values, the externality may yield **equilibrium non-existence**, preventing decentralization

The Rothschild-Stiglitz (1976) Allocation ($\mu = 1$)



Decentralization: Additional Restrictions on Trades

The approach relies of (non-modeled) institutions performing two roles.

1. They monitor the (IC) nature of trades **in each** market

2. They monitor individual trade **across** markets by forcing types to trade in the markets designed for them

A severe control on individuals' aggregate trades is imposed

EEG Scholars: "These assumptions strain the standard view of competitive markets where individual trade is anonymous and the price system alone decentralizes efficient allocations"

Markets and Contracts: Where Do We Stand?

We still lack a comprehensive analysis of the behavior of (competitive) market economies under informational frictions

A) Can we establish welfare theorems <u>in a constrained sense</u> ?

B) How do market outcomes respond to variations in the set of available contracts?

C) Is there a canonical way to represent agents' communication ?

Competitive Screening: Outline

1. Market Breakdown: Akerlof (1970) vs Rothschild-Stiglitz (1976)
2. Nonexclusive Competition under Adverse Selection

Market Breakdown: Akerlof (1970) vs Rothschild-Stiglitz (1976)

Plan of the Lecture

1. A model of trade under adverse selection
2. A benchmark: the competitive screening game under exclusivity.
3. Non-exclusive trades: pooling at equilibrium is a relevant feature of these situations.
4. Discuss extensions and generalizations.

The Environment: Trade under Asymmetric Information

We study the trade between a single seller and a finite number of buyers indexed by $i = 1, 2, ..., K$ with $K \geq 2$.

The seller's endowment is one unit of a perfectly divisible good.

Exchanges are potentially limited by asymmetric information. At the trading stage, the seller is privately informed of the quality of the good.

$\theta \in \Theta$ is the type of the seller.

Payoffs: a Common Value Case

The Seller

$$U(Q, T, \theta) = T - \theta Q,$$

where $Q = \sum_{i=1}^{K} q^i$ and $T = \sum_{i=1}^{K} t^i$ denote aggregate quantities and transfers.

The Buyers

$$V^i(q^i, t^i, \theta) = v(\theta) q^i - t^i.$$

Gains from trade arise if $v(\theta) > \theta$ for some θ.

Competition

Competitive screening: buyers compete in menus for the good offered by the seller.

1. Each buyer k proposes a menu of contracts: $\Gamma^k = \{(q^k, t^k) \in [0,1] \times \mathbb{R}_+\}$ containing at least the no-trade contract $(0,0)$.

2. After learning the quality θ, the seller selects one contract (q^k, t^k) from each of the Γ^i menus, subject to the constraint that $Q \leq 1$.

Exclusive vs Nonexclusive Competition

For a given array of posted menus $\{\Gamma^1, \Gamma^2, \cdots, \Gamma^K\}$:

Exclusive Competition:

The vector $\left(q_k, t_k\right)_{k=1}^{K}$ should contain **at most** one item different from $(0, 0)$.

Nonexclusive Competition:

Any vector $\left(q_k, t_k\right)_{k=1}^{K}$ can be chosen.

Strategies and Equilibrium under Nonexclusivity

A pure strategy for the seller is a mapping associating to each type θ and each profile of menus $(\Gamma^1, \ldots, \Gamma^K)$ a vector

$\big((q^1, t^1), \ldots, (q^K, t^K)\big) \in ([0,1] \times \mathbb{R}_+)^K : (q^k, t^k) \in \Gamma^k$ and $\sum_k q^k \leq 1$.

We take buyers' menus to be compact sets: the seller's problem

$$\sup \left\{ \sum_k t^k - \theta \sum_k q^k : \sum_k q^k \leq 1 \text{ and } (q^k, t^k) \in \Gamma^k \text{ for all } k \right\}$$

has a solution for any type θ and menu profile $(\Gamma^1, \ldots, \Gamma^K)$.

We consider pure strategy *PBE* of the menu game.

Applications

1. Security trading (DeMarzo-Duffie 1999): θ is the value of the asset to the issuer and $v(\theta)$ is the value to the liquidity supplier.
2. Labor market model (Mas-Colell-Whinston-Green 1995): θ is the worker's opportunity cost and $v(\theta)$ is the productivity.
3. Insurance under the dual model of Yaari (1987). If x is the probability of an accident:

$$U = f(x)[W - P + R - L] + \big(1 - f(x)\big)[W - P],$$

Taking $Q = R$, $T = W - P$, $\theta = -f(x)$, and $v(\theta) = -f^{-1}(-\theta)$, we get back to our setting.

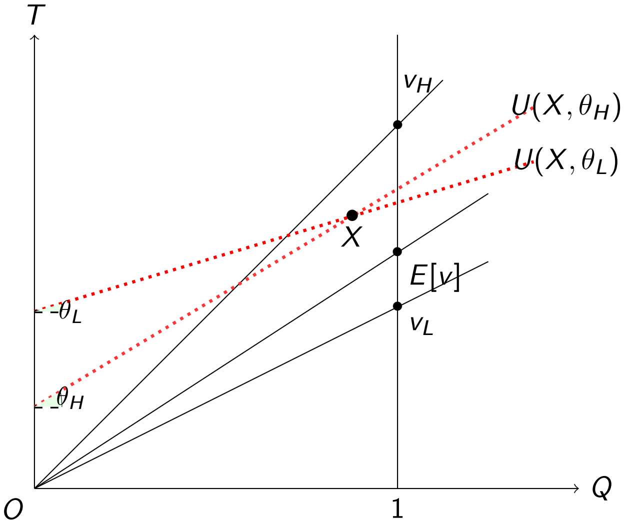
The Two-Type Case

We consider the binary case: $\Theta = \{\theta_L, \theta_H\}$, for some $\theta_H > \theta_L > 0$ and $m = prob(\theta_H)$.

$(A1)$ The seller's and the buyers' perceptions of the quality of the good move together: $v_H > v_L$.

$(A2)$ It would be efficient to trade no matter the quality of the good: $v_L > \theta_L$ and $v_H > \theta_H$.

Feasible (Aggregate) Allocations: (Q, T)



Linear Pricing (Akerlof (1970))

Buyers are <u>restricted to</u> compete through linear menus

A strategy for buyer k is

$$\bar{\Gamma}^k = \{(q^k, t^k) \in [0,1] \times \mathbb{R}_+ : \frac{t^k}{q^k} = p \in (0, +\infty)\}$$

Proposition (Pauly (1974))

Assume (A1) and (A2). Then $\bar{\Gamma}$ always admits an equilibrium:

1. $E(v) > \overline{\theta} \quad \Longrightarrow \quad (Q_L, T_L) = (Q_H, T_H) = (1, E(v))$
2. $E(v) < \overline{\theta} \quad \Longrightarrow \quad (Q_L, T_L) = (1, v_L); (Q_H, T_H) = (0, 0)$

Exclusive Competition (Rothschild-Stiglitz (1976))

A standard competitive screening environment:

Proposition (Rothschild-Stiglitz (1976))

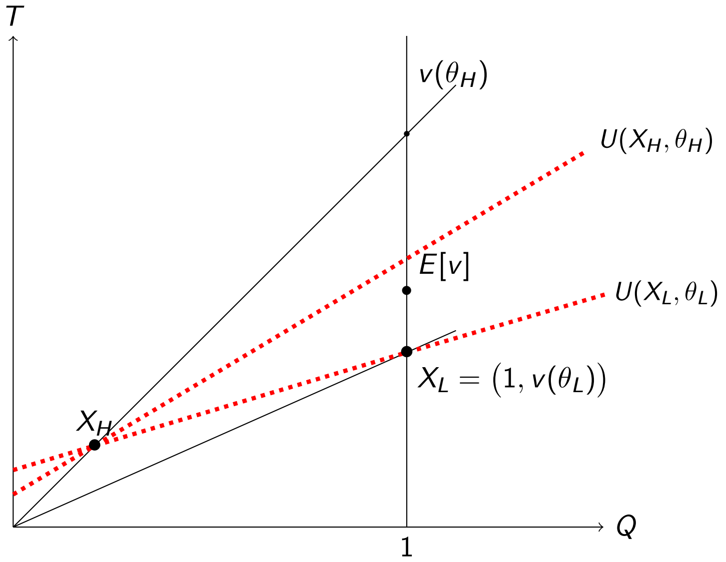
Any equilibrium of Γ^E is separating, with

- $(Q_L, T_L) = (1, v_L)$
- $(Q_H, T_H) = \frac{v_L - \theta_L}{v_H - \theta_L}(1, v_H)$.

Γ^E admits an equilibrium if and only if:

$$m \leq \frac{\theta_H - \theta_L}{v_H - \theta_L}.$$

Exclusivity: Separating Equilibrium



On Market Equilibria under Nonexclusivity

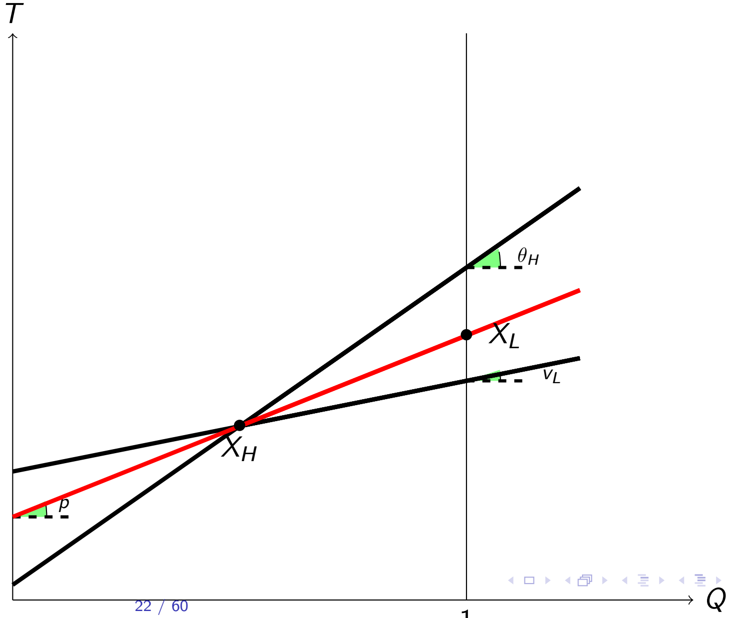
At any separating equilibrium, $Q_H \leq Q_L = 1$.

The allocations X_L and X_H are connected by a line of slope $p \in [v_L, \theta_H]$:

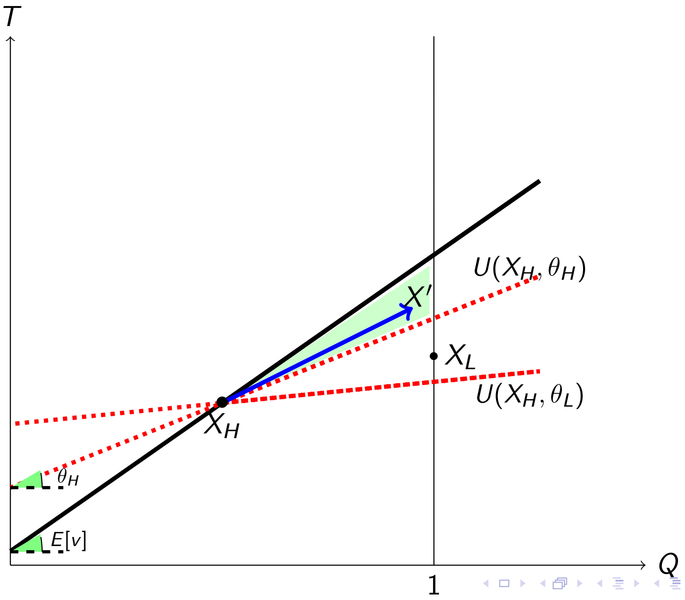
- If $p > \theta_H = MRS_{\theta_H}$, the θ_H seller will trade X_L (violating (IC))
- If $p < v_L$, one can gain proposing to trade $\left(1 - Q_H\right)$ at a price $p' = \theta_L + \varepsilon$.

A high rent is left to the θ_L seller (NOT indifferent between X_L and X_H).

A Candidate Separating Equilibrium



$E[v(\theta)] > \theta_H \implies$ there is no separating equilibrium



Mild Adverse Selection: $E[v(\theta)] > \theta_H$

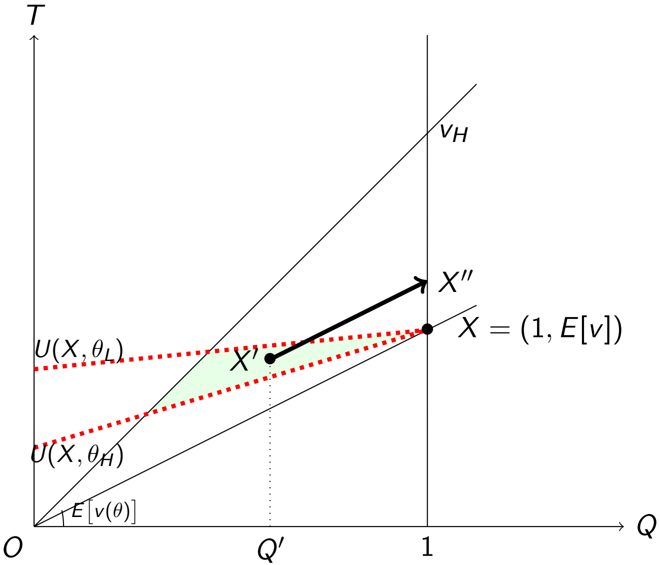
- There is an equilibrium where each buyer offers:

$$\{(q, t) \in [0, 1] \times \mathbb{R}_+ : t = E[v(\theta)]q\},$$

  standing ready to buy any quantity of the good at a constant unit price $E[v(\theta)]$. Both types of the seller set $Q = 1$.

- Any equilibrium allocation is pooling: $(Q_L, T_L) = (Q_H, T_H) = (1, E[v(\theta)])$.

The Linear Price Pooling Equilibrium



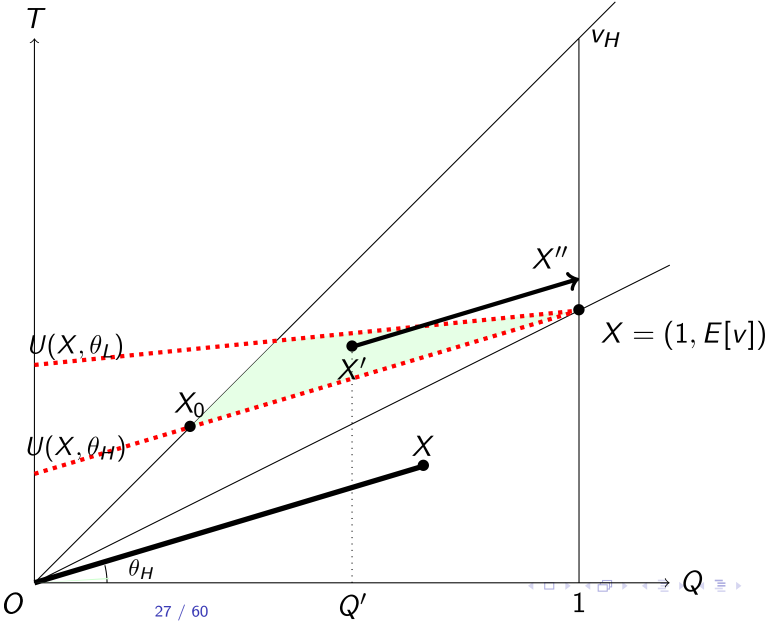
Pooling with Non-Linear Prices: an Example

Linear prices are not a necessary feature of the construction

The allocation $\left(1, E[v]\right)$ can be supported if every buyer offers the menu

$$\left\{(q, t) \in \left[0, \frac{v_H - E[v(\theta)]}{v_H - \theta_H}\right] \times \mathbb{R}_+ : t = \theta_H q\right\} \cup \{(1, E[v(\theta)])\}.$$

Pooling with Non-Linear Prices



Equilibrium Menus

Under mild adverse selection, the equilibrium allocation $(1, E[v(\theta)])$ traded by both types should available if any buyer withdraws his menu offer: **no buyer is indespensable** .

Under mild adverse selection, the unit price of any contract issued in an equilibrium of the non-exclusive competition game is at most $E[v(\theta)]$.

In particular, the unit price of any contract traded in an equilibrium of the non-exclusive competition game is *exactly* $E[v(\theta)]$.

Strong Adverse Selection: $E[v] < \theta_H$

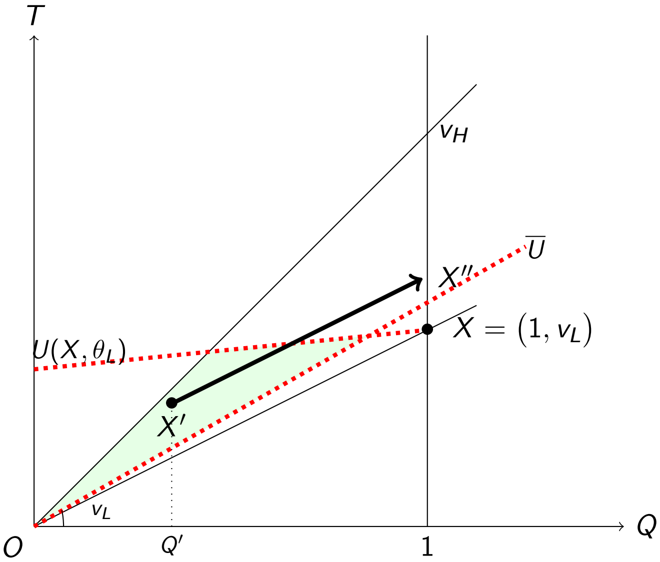
If $E[v(\theta)] < \theta_H$:

There is always an equilibrium in which each buyer offers the menu $\{(q, t) \in [0, 1] \times \mathbb{R}_+ : t = v_L q\}$.

Type θ_L sells $Q = 1$, type θ_H is excluded from trades.

Each equilibrium allocation involves exclusion of θ_H: $(Q_H, T_H) = (0, 0)$ *and* $(Q_L, T_L) = (1, v_L)$.

The Linear Price Separating Equilibrium



Double deviations

Any profitable deviation must involve trading with both types.

The most profitable one leads to trading Q_H at unit price $\overline{\theta}$ with type θ_H, and trading a quantity 1 at unit price $\theta_H Q_H + v_L(1 - Q_H)$ with type θ_L.

The corresponding payoff is:

$$m[v_H - \theta_H]Q_H + (1-m)[v_L - \theta_H Q_H - v_L(1 - Q_H)] = \{E[v(\theta)] - \theta_H\}Q_H,$$

which is at most zero under strong adverse selection.

Equilibrium Menus and Latent Contracts

Under both mild and strong adverse selection and in any perfect Bayesian equilibrium, an infinite number of latent contracts must remain available if any of buyers withdraws his offers.

Remark: The equilibrium allocations cannot be supported if buyers were restricted to the use of simple mechanisms.

The Two-Type Environment: a Resume

1. Pure strategy equilibria always exist.
2. Equilibrium aggregate allocations are unique. They involve either pooling or exclusion of the θ_H type (Akerlof outcome).
3. Equilibria are separating if $m \leq \dfrac{\theta_H - v_L}{v_H - v_L} < m^e = \dfrac{\theta_H - v_L}{v_H - \theta_L}$.

Continuum of Types

The results extend to the continuous-type case.

The seller type $\theta \in [\theta_L, \theta_H]$, with F and $f > 0$.

Continuum of Types: Exclusivity

No allocation can be supported at a pure strategy equilibrium (Riley, 2001).

Any equilibrium allocation must satisfy $U(\theta) = \big(v(\theta) - \theta\big) Q(\theta)$: buyers earn zero profit on each type of sellers.

It is always possible to profitably deviate buying one unit from a (small) interval of types containing θ_L.

Continuum of Types: Non-Exclusivity

EXISTENCE: There always exists a pure strategy PBE in which each buyer proposes to buy any quantity at a unit price $p^* = E\left[v(\theta)|\theta < p^*\right]$.

1. Any deviation (q', t') must involve a price $\frac{t'}{q'} = p' > p^*$.
2. All types $\theta < p^*$ have an incentive to trade the same contract with the deviator.
3. Given the definition of p^*, the deviation is non-profitable.

Continuum of Types: Non-Exclusivity

UNIQUENESS: The aggregate quantity traded at any equilibrium is $Q(\theta) = 1$ if $\theta < p^*$, and $Q(\theta) = 0$ if $\theta > p^*$. Buyers get zero-profits.

1. Suppose there is a $\theta_1 < p^*$ selling $Q_1 < 1$.
2. The deviation $\big((1 - Q_1), \theta_1(1 - Q_1)\big)$ attracts all types below θ_1.
3. The deviation is profitable:

$$(1 - Q_1) \int_{\theta_L}^{\theta_1} (v(\theta) - \theta_1) dF(\theta) > 0$$

Nonexclusive Competition under Adverse Selection

The Trading Environment

Trading takes place between a single seller and a finite number of buyers indexed by $i = 1, \ldots, K$, with $K \geq 2$

A feasible trade is an arbitrary vector $((q^1, t^1), \ldots, (q^K, t^K)) \in \mathbb{R}^{2n}$: there are no a priori restrictions on the sign of transactions

The seller is privately informed of the quality of her good (type) $i, j \in \{L, H\}$; denote by $m_i > 0$ the probability that she is of type i, $m_L + m_H = 1$

Preferences: a General Common Value Case

The type i seller has a utility function $u_i(Q, T)$, where $Q = \sum_k q^k$ and $T = \sum_k t^k$ are aggregate quantities and transfers

$u_i(Q, T)$: differentiable, and strictly quasi-concave in (Q, T)

Seller's preferences satisfy strict single-crossing: $\tau_H(Q, T) > \tau_L(Q, T)$ $\forall$ (Q, T)

$b_i^k = v_i q - t$: payoff to buyer k trading (q, t) with the type i seller, with $v_H > v_L$

$v = m_L v_L + m_H v_H \in (v_L, v_H)$ is the average quality

Applications

Application I: Pure Trade

The seller's utility is quasilinear as in Biais-Martimort-Rochet (2000) and Back-Baruch (2012)

$$u_i(Q, T) = T - c_i(Q), \quad c'_H > c'_L$$

Application II: Insurance

As in Rothschild and Stiglitz (1976), the seller faces a binary lottery on her wealth, $\pi_i W_G \oplus (1 - \pi_i) W_B$, and she has utility

$$\pi_i u(W_G - P) + (1 - \pi_i) u(W_B - P + R)$$

Letting $q \equiv r$, and $t \equiv -p$, and $\pi_H > \pi_L$, we get

$$u_i(Q, T) = \pi_i u(W_G + T) + (1 - \pi_i) u(W_B + Q + T)$$

Application III: Credit

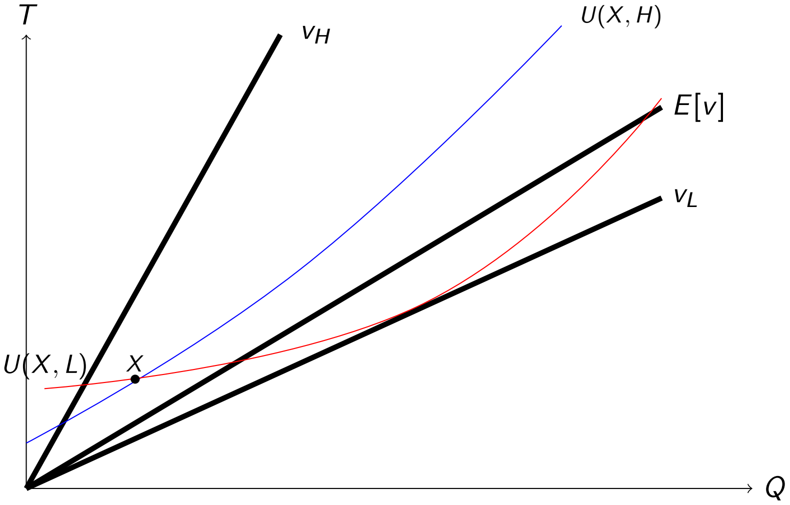
As in Stiglitz-Weiss (1981), borrower raises a capital $B > 0$ from several investors. There are two aggregate states: default and no-default.

The type of the borrower affects both the probability of the no-default state, π_i, and the cash flows in that state, $f_i(B)$

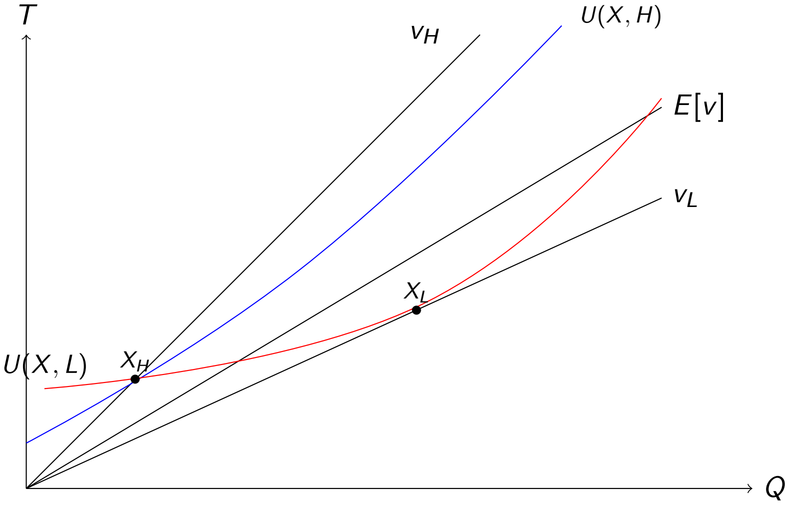
- A (debt) contract is a borrowed capital-promised repayment pair (b, p)
- Each investor's payoff: $\pi_i p - b \iff v_i q - t$ if $v_i \equiv \pi_i$, $q \equiv p$, $t \equiv b$
- Borrower's payoff: $\pi_i[f_i(B) - P] \iff \pi_i[f_i(T) - Q] = u_i(Q, T)$

The borrower raises T against a promise of repaying Q

Aggregate Allocations ($Q_L \geq Q_H \geq 0$)



Exclusivity ($Q_L > Q_H \geq 0$)



Pivoting

Under exclusive competition, what matters from the viewpoint of any given buyer is just the maximum utility that the seller can get by trading with some other buyer

Under non-exclusive competition, all the contracts offered by the other buyers matter from the viewpoint of any given buyer, who can use his competitors' offers to propose attractive deviations

For instance, if (Q^{-k}, T^{-k}) is made available by buyers other than k, buyer k can *pivot on* (Q^{-k}, T^{-k}) *to attract* type i by offering the contract $(Q_i - Q^{-k}, T_i - T^{-k} + \varepsilon)$ for some $\varepsilon > 0$

Pivoting II

Under non-exclusive competition, each buyer k essentially faces a seller whose type is unknown, but whose preferences are defined by the indirect utility function

$$z_i^{-k}(q,t) = \max\left\{u_i\left(q + \sum_{l \neq k} q^l, t + \sum_{l \neq k} t^l\right) : (q^l, t^l) \in C^l \; \forall\, l \neq k\right\}$$

The difficulty stems from the fact that the functions z_i^{-k} are endogenous, since they depend on the menus offered by the buyers other than k, on which we impose little restrictions

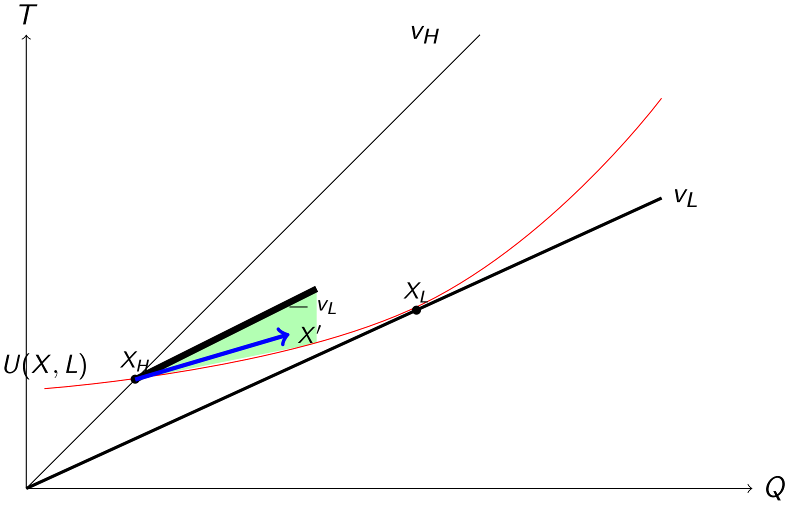
Pivoting III

As a result, there is no a priori guarantee that the functions z_i^{-k} are well behaved: for instance, they could fail to satisfy a single-crossing property, unlike the seller's utility function u_i over aggregate trades

This prevents us from using standard mechanism techniques to characterize each buyer's best response;

Instead, we rely only on pivoting arguments to fully characterize candidate aggregate equilibrium allocations

Pivoting: No Rothschild-Stiglitz Equilibrium



A Basic Result ($Q_L \geq Q_H \geq 0$)

Buyers cannot make profits from trading $(Q_L - Q_H, T_L - T_H)$ with type L

Proposition 1 (Attar-Mariotti-Salanié (2014))

In any equilibrium, $S_L \equiv v_L(Q_L - Q_H) - (T_L - T_H) \leq 0$

<u>Proof</u> (Free-Entry): if $v_L(Q_L - Q_H) > T_L - T_H$, an entrant could pivot on (Q_H, T_H) to attract type L; attracting type H can do no harm since $v_H(Q_L - Q_H) \geq v_L(Q_L - Q_H)$ as $v_H > v_L$ and $Q_L \geq Q_H$

In the insurance model, this rules out Rothschild-Stiglitz (1976) outcome

The Zero Profit Result ($Q_L \geq Q_H \geq 0$)

Proposition 2 (Attar-Mariotti-Salanié (2014))

In any equilibrium, $B = b^k = 0$ for all k

Proof (Free-Entry): if $B_H > 0$, an entrant could pivot on $(0, 0)$ to attract type $H \implies$ this trade must also attract type L, and $vQ_H - T_H \leq 0$:

- $B = vQ_H - T_H + m_L S_L$
- $S_L \leq 0$ by Proposition 1; $vQ_H - T_H \leq 0$ when $B_H > 0$. Then, $B = 0$

The Basic Price Structure of Equilibria

Proposition 3 (Attar-Mariotti-Salanié (2014))

In a pooling equilibrium, $T_L = vQ_L = T_H = vQ_H$*; in a separating equilibrium,*

(i) $T_L = v_L Q_L$ *and* $T_H = v_H Q_H$ *if* $Q_L > 0 > Q_H$

(ii) $T_H = vQ_H$ *and* $S_L = 0$ *if* $Q_L > Q_H \geq 0$

(iii) $T_L = vQ_L$ *and* $S_H = 0$ *if* $0 \geq Q_L > Q_H$

Equilibrium Pricing: case (*ii*)

Everything happens as if both types were selling an aggregate quantity Q_H at unit price v, with type L selling an additional quantity $Q_L - Q_H$ at unit price v_L.

Thus, if $Q_H > 0$, cross-subsidization takes place in equilibrium, with $B_L < 0 < B_H$.

The No Cross-Subsidization Result I

Can there exist equilibria with cross-subsidies, that is, pooling equilibria with non-zero trade, or separating equilibria where both types trade non-zero quantities on the same side of the market?

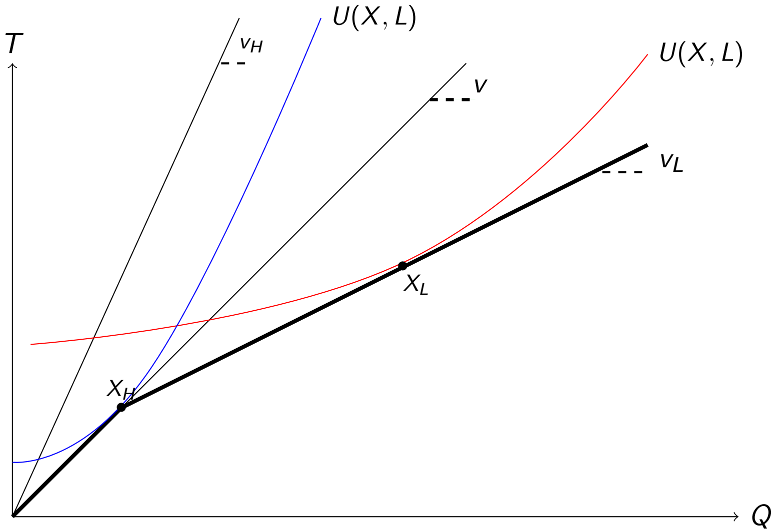
Proposition 4 (Attar-Mariotti-Salanié (2014)

In any equilibrium, $B_L = B_H = 0$

Only Akerlof-like outcomes survive at equilibrium.

Pooling equilibria involve no aggregate trade.

The No Cross-Subsidization Result: $Q_L \geq Q_H \geq 0$



The No Cross-Subsidization Result II: $Q_L \geq Q_H \geq 0$

If buyers make strictly positive aggregate profits when trading with type H, then type H trades efficiently in equilibrium

Lemma 1: *If $B_H > 0$, then $\tau_H(Q_H, T_H) = v$*

If $\tau_H(Q_H, T_H) \neq v$, any buyer could attempt to reap the aggregate profit on type H, while making limited losses on his trades with type L.

For this deviation not to be profitable, it must therefore be that, in equilibrium, the profit that each buyer k makes with type H is no less than the aggregate profit B_H on type H, so that $B_H \leq 0$, a contradiction

The No Cross-Subsidization Result III: $Q_L \geq Q_H \geq 0$

If buyers make strictly positive aggregate profits when trading with type H, the aggregate equilibrium trade of type H remains available if any buyer withdraws his menu

Lemma 2: *If $B_H > 0$ then, for each k, the trade (Q_H, T_H) can be made with the buyers other than k*

If $B_H > 0$, there exists for each k a trade (Q^{-k}, T^{-k}) with the buyers other than k such that $u_H(Q^{-k}, T^{-k}) = U_H$, for, otherwise, buyer k could deviate and reap the aggregate profits on type H.

From Lemma 1, we get that if $Q^{-k} \neq Q_H$, then $T^{-k} > vQ^{-k}$. We finally show that if $Q^{-k} \neq Q_H$, buyer k could profitably deviate by pivoting on (Q^{-k}, T^{-k})

The No Cross-Subsidization Result IV: $Q_L \geq Q_H \geq 0$

Suppose that $B_H > 0$, and let k be such that $b_H^k > 0$.

Buyer k deviates to $\{c_L^k = ((Q_L - Q_H), v_L(Q_L - Q_H)), c_H^k = (q_H^k, t_H^k)\}$.

By Lemma 2, following buyer k's deviation, type H can still trade (Q_H, T_H) in the aggregate

Following buyer k's deviation, type L can still trade (Q_L, T_L) in the aggregate. By perturbing c_L^k , buyer k can make sure that L opts for c_L^k

Buyer k can hence reap all profits on type H of the seller, incurring only a limited loss on type L. **The deviation is profitable**

Equilibrium Aggregate Trades I

Lemma 3: If $Q_L \neq 0$, then $\tau_L(Q_L, T_L) = v_L$

By Proposition 4, $T_L = v_L Q_L$. If $\tau_L(Q_L, T_L) \neq v_L$, there exists a contract offering to buy a strictly positive quantity at a unit price lower than v_L, and that would give type L a strictly higher utility than (Q_L, T_L).

Any of the buyers could profitably attract type L by proposing this contract, which would be even more profitable for the deviating buyer if traded by type H. Hence type L must trade efficiently in equilibrium

Equilibrium Aggregate Trades II

Let Q_L^* be such that $\tau_L(Q_L^*, v_L Q_L^*) = v_L$

Proposition 5 (Attar-Mariotti-Salanié (2014)

If an equilibrium exists, then $\tau_L(0,0) \leq v \leq \tau_H(0,0)$.

- *If* $v_L \leq \tau_L(0,0) \leq v \leq \tau_H(0,0) \leq v_H$, *all equilibria are pooling, with* $Q_L = Q_H = 0$.
- *Otherwise, all equilibria are separating, with* $Q_L = Q_L^* > 0 = Q_H$ *if* $\quad \tau_L(0,0) < v_L, \quad v \leq \tau_H(0,0) \leq v_H$

Separating Equilibria: $Q_L > Q_H \geq 0$